

Formula 1 Forecast: Build Your Grid

Milestone #: 1

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Group Number: 61

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above.

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

Project Description

The project details and way to view potential F1 odds, and gain intel about those odds through provided information. The user can also create an account and view others who are also using the program. The program offers a point system which the user can interact with as they watch their predictions.

Language

SQL and JavaScript

Changes

No changes suggested

An explicit acknowledgment about your use of AI tools in this assignment. Specifically, We are looking for a clear yes/no about whether you have used one or more AI tools. If yes, we want to know which tool(s) you have used and we want a PDF of the full Conversation. No generative tools were used

INSERT:

controllers/[UserController.js](#)

Line 23.

Note: It affects the Scores table as well, as you can see with the insert function above

UPDATE:

controllers/[UserController.js](#)

Line 144

DELETE:

controllers/[UserController.js](#)

Line 194

SELECT

controllers/[UserController.js](#)

Line 135

PROJECTION

controllers/[oddsCategoryController.js](#)

Line 47

JOIN

prediction-engine/[raceodds.js](#)

Line 14

GROUP BY

prediction-engine/[sessionController.js](#)

Line 21

GROUP BY HAVING

prediction-engine/[sessionController.js](#)

Line 33

Having:

```
SELECT FIRSTNAME || ' ' || LASTNAME AS FULLNAME, COUNT(*) AS PODIUMS
  FROM RACE_RESULT NATURAL JOIN DRIVER
 WHERE SEASON=${season} AND POSITION <= 3
  GROUP BY DRIVERID, FIRSTNAME, LASTNAME
  HAVING COUNT(*) >= 2
  ORDER BY PODIUMS DESC
```

Select only the drivers who have placed <= 3. Then we group by the driver name, and only choose resulting tuples with at least >= 2 entries. This represents the number of podiums

GroupBy:

```
`SELECT nationality,
        AVG(accumulatedpoints) AS avg_points
FROM DRIVER
GROUP BY nationality`
```

We can combine drivers with a single nationality and the avg over all of their accumulated points to get average points per nationality.

Group by Aggregation:

```
`SELECT teamid, season, AVG(avg_position) AS avg_finish_position
FROM (
    SELECT teamid, season, trackname, AVG(position) AS avg_position
    FROM RACE_RESULT
    GROUP BY teamid, season, trackname
)
GROUP BY teamid, season
ORDER BY season, avg_finish_position`
```

Select the avg position of all races (per teamid, season, trackname) in the inner from query. Then we take that and group by the season and keep averaging the already averaged position to get the final value. Making it per team, then per season.

Changes to schema:

- Result was broken up into an ISA with Race_Session, Qualifying_Session, Sprint_Session. Each will now reference a team and a session in addition to the driver. This is necessary so we can differentiate between different F1 results, because they are all fundamentally different types of races and require different result data.
- Predictions no longer reference a result. This is wrong because we are often referencing a Result that does not exist, which would require us to create a dummy Result before the actual results are created. This is unnecessarily complicated. Predictions now references a Driver, Team, and Race_Session
- The relationship TakePart was removed because it complicates the F1 data fetching
- We denormalized the Driver and the Team because it has too few attributes to make normalization necessary.

Visual db representation

Please note that the data in our tuples depends on actual f1 data. The below values pertaining to f1 related tuples are not present in the sql script and need to be added via the frontend by clicking a button. The tuples there are just to illustrate the relationships

SCORE

ranking	acc	amount	deductions
1	Aliceacc	100	0
2	Bobacc	80	5
3	Charlieacc	60	10
3	Danaacc	60	10

APP_USER

app_userid	dateoffirstp	user_name	streak	password	acc
Alice!userid	2026-01-01	Alice	3	1234	Aliceacc
Bob!userid	2026-02-15	Bob	5	2345	Bobacc
Charlie!use	2026-03-10	Charlie	2	3456	Charlieacc
Dana!userid	2026-03-20	Dana	1	4567	Danaacc

FRIEND

user1id	user2id			
Alice!userid	Bob!userid			
Bob!userid	Charlie!userid			
Charlie!userid	Alice!userid			
Dana!userid	Bob!userid			
Dana!userid	Alice!userid			

CATEGORY

categoryid	name			
driverodds	Odds for a given driver to win the race			
teamraceodds	Odds for a given team to accumulate the most			
podiumodds	Odds for a driver with podium finishes this season			

PREDICTION

predictionid	categoryid	prediction_	odds_value	date_filed	time_filed	season	trackname	driverid
P01	Driverodds	UNDR	6.0	2026-01-01	2026-01-01	2026	race_track1	A1
These values here depend on the predictions the user makes								

TEAM

points	name	teamid	nationality	
135	Demo1	demo_1	Canadian	
120	Demo2	demo_2	American	
The state of this table depends on current F1 data				

DRIVER

driverid	accumulate	drivernumk	firstname	lastname	nationality	teamid	dateofbirth
A1	60	2	John	Doe	Canadian	demo_1	2026-01-01
S2	50	1	Jane	Doe	American	demo_2	2026-01-01
The state of this table depends on current F1 data							

RACE_SESSION

season	trackname	sessiondate		
demo1	race_track1	2026-01-01		
demo2	race_track2	2026-01-01		
The state of this table depends on current F1 data				

RACE

season	trackname			
2026	race_track1			
2026	race_track2			
The state of this table depends on current F1 data				

SPRINT

season	trackname			
2026	race_track1			
2026	race_track2			
The state of this table depends on current F1 data				

QUALIFYING

season	trackname	qualifyingdate	
2026	race_track1	2026-01-01	
2026	race_track2	2026-01-01	
The state of this table depends on current F1 data			

PRACTICE

SPRINT_RESULT

[illegible]